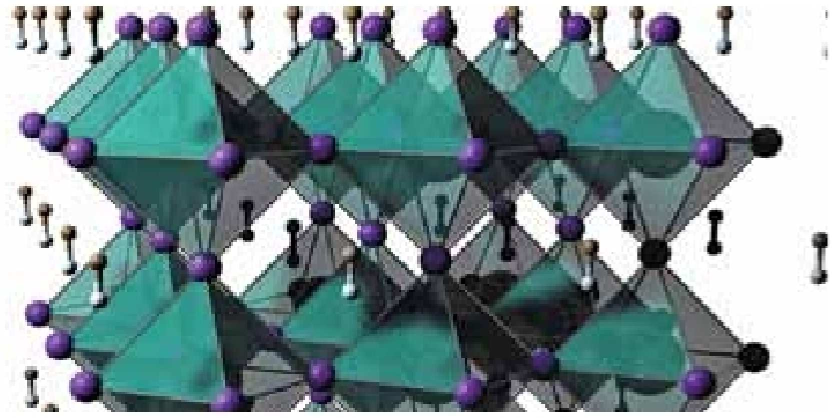
Key advance for printing circuitry on wearable fabrics

Much effort has gone into integrating sensors, displays, power sources, and logic circuits into various fabrics for the creation of wearable electronic textiles. Fabricating rigid devices on cloth, which has a surface that's both porous and non-uniform, is tedious and expensive, requiring a lot of heat and energy. It also limits the flexibility and wearability of the fabric.

Now, a stable, printable ink, based on binary metal iodide salts, thermally transforms into a dense compound of cesium, tin, and iodine. The resulting film of Cs_2SnI_6 has a crystal structure that makes it a perovskite.

Thanks to the perovskite film, negative-temperature-coefficient thermistors can be directly woven onto polyester at temperatures as low as 120 degrees Celsius—just 20 degrees higher than the boiling point of water. For the future of wearable e-textiles, the ink and the thermistor performance are all promising.



Perovskite crystal structure (Credit: https://today.oregonstate.edu)

EV batteries can now be charged up to 90% in just 6 minutes

A POSTECH research team has developed a faster charging and longer lasting battery material for electric vehicles (EV). For fast charging and discharging of Li-ion batteries, methods that reduce the particle size of electrode materials were used. However, a reduced particle size decreases the volumetric energy density of the batteries.

To this, the research team confirmed that if an intermediate phase in the phase transition is formed during the charging and discharging, high power can be generated without losing high energy density or reducing the particle size through rapid charging and discharging, enabling the development of long-lasting Li-ion batteries.

This in turn proved that the intermediate phase formation can dramatically increase the charging and discharging speed of the cell by creating a homogenous electrochemical reaction in the electrode. As a result, the synthesised Li-ion battery electrodes charge up to 90% in six minutes and discharge 54% in 18 seconds, a promising sign for developing high-power Li-ion batteries.

Robot remotely checks Covid-19 patients' vital signs

During the current coronavirus pandemic, one of the riskiest parts of a healthcare worker's job is assessing people who have symptoms of Covid-19. Researchers from MIT, Brigham, and Women's Hospital hope to reduce that risk by using robots to remotely check patients' vital signs. Controlled by a handheld device, these can also carry a tablet PC that allows doctors to ask patients about their symptoms without being in the same room.

Researchers from MIT and Brigham and Women's Hospital hope to reduce the risk to healthcare workers posed by Covid-19 by using robots to remotely measure patients' vital signs (Credit: Spot image courtesy of the researchers, edited by MIT News)

Using four cameras—an infrared camera plus three monochrome cameras—that filter different wavelengths of light mounted on a dog-like robot developed by Boston Dynamics, the researchers have shown that they can measure skin temperature, breathing rate, pulse rate, and blood oxygen saturation.

For body temperature, the camera measures skin temperature on the face, and the algorithm correlates that temperature with core body temperature. As the patient breathes in and out, wearing a mask, their breath changes the temperature of the mask. Measuring this temperature change allows the researchers to calculate how rapidly the patient is breathing.

The three monochrome cameras allow the researchers to measure the slight colour changes that result when hemoglobin in blood cells binds to oxygen and flows through blood vessels. An algorithm uses these measurements to calculate both pulse rate and blood oxygen saturation.